

**Projects: Generative AI-Based Ligand Binding Pose
Refinement from Physics-Based Priors**

**Literature Review: Scoring Functions for Ligand Binding Pose
Prediction**

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Abstract

Predicting how a small molecule binds to a protein is a problem in structure-based drug design and the scoring function used to judge a candidate pose has a direct effect on how reliable that prediction is. This review looks at five categories of scoring functions used in docking and pose prediction: empirical, knowledge-based, machine-learning-based and generative AI-based methods. For each category the underlying mechanism is described representative methods are. The reported strengths and weaknesses are summarized, with particular attention paid to computational cost, interpretability and how well each approach accounts for conformational entropy and desolvation two effects that remain poorly handled across nearly all classical scoring methods. A consistent pattern emerges across the review: machine-learning and generative models tend to score on raw affinity prediction and produce faster more flexible pose samples but they do not reliably outperform physics-based methods when it comes to selecting the correct pose and generative models in particular can output poses that appear accurate by RMSD while still being chemically implausible. Based on this comparison the review concludes that empirical scoring functions, AutoDock Vina and its Vinardo variant are the most suitable choice as a physics-based prior for a generative pose-refinement pipeline. This is due to their speed, physical grounding and established use across the docking literature along with a well-documented weakness, in modeling entropy that gives a downstream refinement model a clear and physically motivated problem to correct.

Review

Force Field Scoring Function:

They are centered on molecular mechanical calculations. These scoring functions rely on the fundamental molecular physics terms such as Van der Waals interactions, electrostatic interactions and desolvation energies. These terms can be derived from both experimental data and ab initio quantum mechanical calculations. Programs such as GoldScore, DOCK and early versions of AutoDock use this type of scoring function.

Pros:

1. Each term in the energy equation relates to a clear physical force, making it easy to understand why a molecule scores well or poorly.
2. Provides a natural solution signal to regularize a generative model, as it doesn't rely on any memorized structural patterns or prior data.

Cons:

1. Computationally more expensive due the solvation and entropy terms.
2. Estimating the loss of conformational freedom when a molecule binds, is very hard and often left out or simplified.
3. Accurately measuring how water molecules leave the binding site remains a major challenge.

Empirical Scoring Function:

They estimate the binding affinity of a complex on the basis of a set of weighted energy terms

$$\Delta G = \sum_i W_i \Delta G_i$$

where ΔG_i represents different energy terms such as VDW energy, electrostatics, hydrogen bond, desolvation, entropy, hydrophobicity, etc. The corresponding coefficients W_i are determined by fitting the binding affinity data of a training set of protein-ligand complexes with known 3D structures. AutoDock Vina is the most widely used example. It evaluates the binding energy using terms for Van der waals forces and hydrogen bonding and runs roughly two orders of magnitude faster than AutoDock4 while beating its accuracy. With the rapid increase in the number of protein-ligand complexes with known 3D structures and binding affinities, it is possible to develop a relatively general empirical scoring function by training with known binding constants of thousands of diverse protein-ligand complexes.

Pros:

1. Scriptable/accessible implementations (Vina, Smina) make integration into custom pipelines realistic.
2. Fast enough to be used repeatedly inside an optimization loop or, plausibly, as a guidance signal during iterative generative sampling.
3. Widely adopted as the field's default baseline, so results are directly comparable across papers.

Cons:

1. One paper I found particularly important pointed out a real weakness in how these functions get built. The standard method of fitting scoring functions using linear regression to maximize correlation with experimental affinities does not actually produce a function that is optimal at the docking task (i.e., distinguishing correct poses from incorrect ones). A function can correlate well with binding affinity on crystal structures yet still be mediocre at telling a near-native pose from a wrong one.
2. The same paper found that training via a combination of scoring, minimization, and re-docking (rather than pure regression) produced a

noticeably better-performing variant (called Vinardo), which outperformed standard Vina across every benchmark tested.

3. Even the best empirical scoring functions (Vina score, ChemPLP, Glide score) still carry a high false-positive rate in virtual screening settings, according to comparative reviews.

Knowledge-Based Scoring Functions:

Knowledge-Based scoring functions (also known as statistical-potential scoring functions) employ energy potentials that are derived from the structural information embedded in experimentally determined atomic structures. Pairwise potentials are directly obtained from the occurrence frequency of atom pairs in a database using the inverse Boltzmann relation.

$$w(r) = -k_B T \ln[g(r)], g(r) = \frac{p(r)}{p^i(r)}$$

here k_B is the Boltzmann constant, T is the absolute temperature, of the system, $p(r)$ is the number density of the protein-ligand atom pair at a distance r , and $p^i(r)$ is the pair density in a reference state where the interatomic interactions are zero. The inverse Boltzmann method provides a simple and effective alternative method to derive the interaction scores from the structural information for complicate protein systems. Compared to the force field and empirical scoring functions, the knowledge-based scoring functions offer a good balance between accuracy and speed. Because the potentials are extracted from the structures rather than from attempting to reproduce the known affinities by fitting, and because the training structural database can be large and diverse, the knowledge-based scoring functions are quite robust and relatively insensitive to the training set. Most of the current knowledge-based scoring functions approximate the reference state with an atom-randomized state by ignoring the effects of excluded volume, interatomic connectivity etc. The scoring function consists of a distance-dependent pair-potential term and a surface-dependent singlet-potential term.

Pros:

1. Computationally cheap once the statistical potentials are precomputed.

Cons:

1. Entirely dependent on the coverage and quality of the structural database they're built from.
2. Less interpretable than force-field methods.

Machine-Learning Scoring Function:

With the rapid progress of computational power and exponential increase of data, ML has been applied in many branches of CADD, such as chemical space exploration, molecular property prediction, protein structure prediction and VS. This scoring function uses supervised learning to learn from the structural data of protein/ligand and their measured binding affinities. Early efforts used traditional ML methods, such as SVM, RF and GBT, to improve scoring performance on benchmark test sets. Their inputs were manually designed descriptors, such as molecular interaction fingerprints, ligand features, atom-pairwise terms and force field terms. RF-score, the first ML scoring function to outperform classical scoring functions on scoring tasks, was proposed by Ballester and Mitchell in 2010. It utilized the random forest algorithm with feature selection comprised of protein–ligand atom-wise pair counts at a predefined distance cutoff. In 2013, Zilian proposed SFCscore which utilized random forest algorithm but with feature selection of 63 empirical terms. Few examples of the terms are rotatable bonds, hydrogen bonds, aromatic interactions and polar contact surfaces. However, both RF-score and SFCscore performed much worse on docking and screening tasks compared to classical scoring functions. In 2017, the Zhang group employed a Δ -machine learning approach, in which a ML model was employed to parametrize corrections to the Vina score. This strategy enabled the scoring function to have both the excellent docking power of Vina and the accurate scoring performance of the ML method. In their work, $\Delta_{\text{vina}}\text{RF}_{20}$ was developed using random forest with feature selection comprised of 10 features related to pharmacophore-based solvent-accessible surface area (SASA) and 10 empirical terms from Vina 58 features. As a result, $\Delta_{\text{vina}}\text{RF}_{20}$ achieved the best performance among a panel of classical scoring functions in all evaluation metrics (scoring, ranking, docking and screening powers) for the CASF 2007 and 2013 benchmarks. In 2017, Wallach and co-workers introduced AtomNet, the first CNN-based scoring function incorporating 3D structural information. The inputs of AtomNet used vectorized 3D grids placed over the protein–ligand interaction

interface, with each grid cell storing a value describing the presence of some basic structural features, varying from a simple enumeration of atom types to more complex descriptors. AtomNet achieved much better AUC value than Vina (classical scoring function) on the DUD-E test set. Several similar CNN-based scoring functions, such as the CNN model proposed by Ragoza et al. (2017). One of the limitations of these CNN models was the dependency on the coordinate frame. Different orientations of the same protein–ligand structure could generate different representations. In order to address this issue, Zheng and co-workers introduced OnionNet in 2019. This method used a CNN model with inputs based on rotation-free element-pair specific contacts between protein and ligand in different shells. OnionNet, as well as the subsequent OnionNet-2.0 (2021), achieved excellent scoring performance on the CASF-2016 benchmark.

Generative AI Based Models:

Rather than scoring a set of candidate poses generated by a separate search algorithm, these models directly generate the ligand pose. EquiBind and TankBind use regression to predict conformations quickly; DiffDock uses a diffusion process over ligand translation, rotation, and torsion angles, paired with a confidence model that ranks the sampled poses at the end.

Pros:

1. Much faster than exhaustive search-based docking for large-scale scenarios.
2. Diffusion-based approaches in particular can represent multimodal pose distributions rather than committing to a single best guess early.
3. Newer variants (SigmaDock, SurfDock) improve on structural realism by decomposing ligands into rigid fragments or incorporating protein surface features directly into the diffusion process.

Cons:

1. The single biggest concern I found across multiple papers: strong RMSD accuracy does not guarantee physical plausibility. It is entirely possible for a generative model to output a pose that is geometrically "close" to the crystal structure by RMSD, yet chemically impossible.

2. Because of RMSD and physical plausibility no DL docking method actually never outperforms any of the physics-based methods (AutoDock Vina or GOLD).

Evaluation Methodology Notes

One thing I kept running into while reading was that how these functions get evaluated matters as much as the functions themselves. PoseBusters was the paper that reframed this for me: it isn't enough to report RMSD, you have to separately check physical validity (no steric clashes, correct bond geometry, sensible stereochemistry). Interestingly, that same study found that post-docking energy minimization significantly increased the number of physically plausible structures for DiffDock, DeepDock and TankBind but did not improve AutoDock Vina or GOLD, because those method's outputs were already physically constrained by the physics-based scoring function driving their search in the first place. That result alone is close to a direct empirical justification for a "physics-based prior" refinement project: generative models clearly benefit from physics correction in a way that already-physics-grounded methods don't need.

Conclusion: Which Scoring Function Is "Best" and Why

After going through all of these, I don't think there's a single function that wins on every axis but for the specific purpose of serving as a physics-based prior inside a generative pose-refinement pipeline, I keep landing on the empirical scoring function (AutoDock, Vinardo) used alongside lightweight force-field terms. My reasoning:

1. Technically ML/DL scoring functions achieve higher correlation with experimental binding affinities on crystal structures. But the literature is fairly consistent that this advantage doesn't carry over to the actual pose-discrimination task, where Vina, ChemPLP functions still hold their own or win outright. Since my project is about pose refinement, not affinity prediction, this is the more relevant benchmark to weight heavily.
2. Nearly every generative docking paper I read like PoseBusters, DiffDock they either use benchmarks against Vina or use Vina-family scores as a part of their evaluation. Using Vina for this project will keep the results comparable to the

existing literature rather than requiring readers to trust an unfamiliar baseline.

3. Force-field-style post-processing measurably improved the physical plausibility of poses generated by diffusion and regression models.
4. Vina is fast and scriptable, it's a genuinely feasible choice for repeated evaluation during iterative or guided generation.
5. Vina's original affinity-regression training approach isn't actually optimized for the docking task it's used for in practice and that retraining with a docking-aware objective (scoring + minimization + re-docking) meaningfully improved performance.

Entropic RNOT's docking-pose-refinement experiment uses GNINA poses scored with Vinardo specifically as its "physical prior" input, for the same architectural reason you're describing: fast, physics-based, good enough to be close, not accurate enough to be right.

Force-field scoring is the more "pure" physics option but is generally too slow to generate large volumes of prior poses for training, and per the citation above, it's the category where the entropy problem is documented as worst, not best. Knowledge-based and ML-based scoring both fold entropy into a fitted statistical shortcut rather than a physical term, which is fine for their own use case (fast ranking) but doesn't give you a "prior" whose errors are physically interpretable. If your refinement model is meant to learn why the prior is wrong and correct it systematically, you want the prior's failure mode to be a known, documented physical gap (missing entropy/desolvation terms).

Empirical/semi-empirical scoring (Vina-family) generates the prior our generative model refines. The generative AI model's whole job is to learn the correction that force-field and empirical scoring structurally cannot make because, per the sources above, none of the four classical categories have a real entropy term, they either omit it or approximate it with a fitted constant.

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